Deepak

Leetcode GFG Github
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Education

Indian Institute Of Technology (IIT), Bhilai
B.Tech in Data Science & Artificial Intelligence *October 2022 - May 2026*

Technical Skills

Languages: Python, SQL, C++
Machine Learning: Feature Engineering, Supervised Learning, Deep Learning, CV, NLP
Generative AI & LLMs: LLMs, Fine-tuning, RAG, AI Agents
LLM Frameworks: Hugging Face, LangChain, LangGraph
Deep Learning Frameworks: PyTorch, TensorFlow
Data Analysis: Pandas, NumPy, EDA, Matplotlib
Databases & Vector Stores: MySQL, MongoDB, Qdrant
MLOps & Deployment: Docker, FastAPI, REST APIs, Streamlit

Experience

Project Engineer – Tribal Research & Innovation Park (TRIP), IIT Bhilai **Aug 2026 – Present**

- Developing **AI/ML solutions** focused on **Generative AI, NLP, and Computer Vision**.
- Integrating **AI/ML models** with **Python, REST APIs, and databases**.
- Supporting **evaluation, deployment, and troubleshooting** of AI/ML applications.

Internship

Machine Learning Research Intern – IIT Bhilai | Certificate **Feb 2025 – Jun 2025**

- Designed and implemented an **end-to-end legal-domain AI chatbot** using curated **Indian legal datasets**.
- Built scalable **data scraping and preprocessing pipelines** to convert large volumes of unstructured legal text into structured **Question–Answer (QA)** datasets using **Python** and **JSON**.
- Applied **parameter-efficient fine-tuning techniques** to adapt language models for legal question answering.
- Focused on **training efficiency and reliability** using checkpointing and memory-aware optimization.
- Developed and deployed an interactive **Streamlit-based web application** to serve real-time legal queries, enabling practical evaluation of the trained chatbot.

Data Science Intern – Encryptix | Certificate **May 2024 – Jun 2024**

- Performed data cleaning, preprocessing, and **EDA** using **Python** and **Pandas** to support business analysis.
- Assisted in building and evaluating baseline models to analyze trends and patterns in datasets.
- Created visual reports using **Matplotlib** and **Seaborn** to communicate insights to non-technical stakeholders.
- Documented findings and observations to support data-driven decision-making and process improvements.

Projects

Multilingual DPR Retrieval System with LLM-Enhanced Hard Negatives | *PyTorch, FAISS, Ollama* **GitHub**

- Built **Dense Passage Retrieval system** with **LLM-powered negative classification** (87.5% accuracy) and **RAG-augmented ranking**, processing 1000+ multilingual samples with 50% reduction in false negatives.
- Implemented **FAISS-indexed retrieval pipeline**, enabling **cross-lingual semantic search** across 9+ languages.
- Achieved **49.4% MRR@10 improvement** through progressive enhancement, with strong multilingual transfer **(19% zero-shot gain)** validated on 15+ languages

AI-Powered Meeting Intelligence Agent | *Python* **Demo Video** **GitHub**

- Built an AI-powered meeting analysis system using **Streamlit**, **FAISS**, and **Groq LLM** for automated summaries and action item extraction.
- Implemented a **RAG** pipeline with sentence-transformers and FAISS for semantic search over meeting transcripts.
- Enabled natural language Q&A using top-k vector similarity retrieval for contextual responses.
- Designed a modular backend handling video uploads, transcript processing, and structured NLP-based insight generation.

Invoice Extraction Using LLaMA 3.2 Vision Model | *Python* **Deployment** **GitHub**

- Developed an AI-powered invoice extraction system using **LLaMA 3.2 Vision Model** and **Groq API**.
- Built a Streamlit-based web app that processes invoice images and exports structured data in JSON & CSV formats.
- Designed robust data validation pipelines using Pydantic, ensuring clean and correctly formatted invoice details.
- Implemented batch processing for multiple invoices via ZIP file uploads, enhancing scalability and efficiency.
- Successfully deployed the application on Render, making it accessible online for real-world usage.

Positions of Responsibility

- **Head, Student Suggestion Forum – IIT Bhilai:** Coordinated student issues for **800–1,000** students
- **Head, Management Committee – IIT Bhilai:** Led a team of **40–50** members and supported college events.
- **Convenor, Meraz 5.0 Informals – IIT Bhilai:** Managed informal events and logistics for a fest with **9,000+ attendees**.

Extracurriculars

- **Student Volunteer :** Student Mentorship Program (SMP) IIT Bhilai
- **Outreach member – CoSA 2024–25, IIT Bhilai**